

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416063
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lincoln; Kyle

Leaching fluid control systems for mining operations

Abstract

The present disclosure provides a leaching fluid control system for mining operations that includes controllable valve circuitry in fluid communication with a leaching fluid source, the controllable valve circuitry to provide a controlled flow rate of the leaching fluid; flow rate sensor circuitry in fluid communication with the controllable valve circuitry and with one or more leaching fluid lines to generate a flow rate signal indicative of the leaching fluid flow rate through the controllable valve circuitry; and controller circuitry to control the controllable valve circuitry to deliver a flow rate of leaching fluid based on the flow rate signal and one or more flow rate thresholds.

Inventors:	Lincoln; Kyle (Wilcox, AZ)
Applicant:	Southwest Irrigation LLC (Wilcox, AZ)
Family ID:	1000008819389
Assignee:	Southwest Leaching Technologies, LLC (Pearce, AZ)
Appl. No.:	18/690689
Filed (or PCT Filed):	September 08, 2022
PCT No.:	PCT/US2022/042965
PCT Pub. No.:	WO2023/039109
PCT Pub. Date:	March 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240271246 A1	Aug. 15, 2024

Related U.S. Application Data

Publication Classification**Int. Cl.:** C22B3/02 (20060101)**U.S. Cl.:**

CPC C22B3/02 (20130101);

Field of Classification Search**CPC:** C22B (3/02); E21B (43/28)**USPC:** 266/78; 266/101; 266/114; 75/744; 75/739; 75/711; 75/723; 75/741; 75/736; 75/733

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1244465	12/1916	Braly	N/A	N/A
1454864	12/1922	Schaake	N/A	N/A
1671706	12/1927	William	N/A	N/A
1833704	12/1930	Bins	N/A	N/A
2721050	12/1954	Sams	N/A	N/A
2804095	12/1956	Hans	N/A	N/A
3776141	12/1972	Gelhard	N/A	N/A
3993293	12/1975	Mukae	324/438	C22B 3/02
4017309	12/1976	Johnson	N/A	N/A
4069765	12/1977	Muller	N/A	N/A
4091070	12/1977	Riggs	423/41	C22B 15/0071
4248628	12/1980	Esna-Ashari	75/724	C22B 60/0226
4625631	12/1985	Vera	N/A	N/A
4753484	12/1987	Stolarczyk et al.	N/A	N/A
5005606	12/1990	Carroll	N/A	N/A
5005806	12/1990	Krauth	N/A	N/A
5226456	12/1992	Semak	N/A	N/A
5240503	12/1992	Levy et al.	N/A	N/A
5256002	12/1992	Shimizu	N/A	N/A
5655457	12/1996	Sherman	N/A	N/A
5901361	12/1998	Luong	N/A	N/A
6360669	12/2001	Albrich	N/A	N/A
6508608	12/2002	Nilson et al.	N/A	N/A
6655641	12/2002	Sherman	N/A	N/A
8202411	12/2011	Buschmann	205/348	C25C 1/22
8801327	12/2013	Johnson	N/A	N/A
10189484	12/2018	Grip	N/A	N/A
10604163	12/2019	Souchal	N/A	N/A
11639539	12/2022	Dean, Jr.	75/743	G08B 21/182

11993826	12/2023	Buffington	N/A	C22B 11/04
2002/0041840	12/2001	Arroyo et al.	N/A	N/A
2003/0000414	12/2002	Sherman	N/A	N/A
2008/0121399	12/2007	Zupanick	N/A	N/A
2013/0092704	12/2012	Tincher et al.	N/A	N/A
2013/0125709	12/2012	Morrow et al.	N/A	N/A
2015/0346374	12/2014	Campbell	N/A	N/A
2019/0041016	12/2018	Glennester	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
203389413	12/2013	CN	N/A

OTHER PUBLICATIONS

CN203389413, mining water filtering system, Ye, Yuan-Yong (Year: 2014). cited by examiner
International Search Report and Written Opinion from related matter PCT/US22/42965 mailed Jan. 4, 2023. cited by applicant

Bester et al. "Reconciliation of the mining value chain-mine to design as a critical enabler for optimal and safe extraction of the mineral reserve." Journal of the Southern African Institute of Mining and Metallurgy 116.5 (2016): 407-411. May 2016. cited by applicant

International Search Report and Written Opinion from related PCT PCT/US22/32368 mailed Oct. 12, 2022. cited by applicant

International Preliminary Report on Patentability from corresponding PCT Appln. No. PCT/US2022/032368, dated Nov. 21, 2023. 6 pages. cited by applicant

Notice of Allowance from related U.S. Appl. No. 18/533,539 mailed Mar. 14, 2024. cited by applicant

International Preliminary Report on Patentability from corresponding PCT Appln. No. PCT/US2022/042965, dated Mar. 5, 2024. 6 pages. cited by applicant

Primary Examiner: Roe; Jessee R

Assistant Examiner: Aboagye; Michael

Attorney, Agent or Firm: Grossman, Tucker, Perreault & Pflieger, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of International application No. PCT/US22/42965 filed 8 Sep. 2022, which is based on and claims the benefit of U.S. Provisional Patent Application Ser. No. 63/241,929, filed 8 Sep. 2021, both of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to leaching fluid control systems for mining operations.

BACKGROUND

(2) Open-pit mines are in operation throughout the world to enable mining of precious and/or industrial minerals such as copper, gold, silver, platinum, etc. An open-pit mine is usually formed as earth material is removed, forming a vast hole. The removed earth is placed on a liner and levelled to form a leach pad. As additional earth material is added to the leach pad, each layer adds to the height of the leach pad, and forms a slope on the sides. The slope section generally runs from

the top of the leach pad (e.g., surface level) to the bottom of the leach pad, and depending on the depth and size of the leach pad, the slope section can be several hundred feet or more in length from top to bottom. To extract minerals from the earth material on the leach pad, a leaching fluid (typically, an acidic solution) is applied to the leach pad to soak through the layers of earth, thus forming a mineral-rich slurry. The slurry may then be collected from the liner, and the dissolved minerals may be recovered from the slurry using known recovery processes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features and advantages of various embodiments of the claimed subject matter will become apparent as the following Detailed Description proceeds, and upon reference to the Drawings, wherein like numerals designate like parts, and in which:
- (2) FIG. 1 illustrates a leaching fluid control system according to various embodiments of the present disclosure;
- (3) FIGS. 2A and 2B illustrate views of a housing structure to contain the flow control system and/or flush control system according to one embodiment of the present disclosure;
- (4) FIGS. 3A and 3B illustrate views of a housing structure to contain the flow control system and/or flush control system according to another embodiment of the present disclosure;
- (5) FIGS. 4A and 4B illustrate views of a leaching pad according to various embodiments of the present disclosure;
- (6) FIG. 5A illustrates the fluid control system deployed on a generally horizontal section of a leaching pad according to one embodiment of the present disclosure;
- (7) FIG. 5B illustrates the fluid control system deployed on the slope section of a leaching pad according to one embodiment of the present disclosure; and
- (8) FIG. 6 illustrates a flowchart of leaching fluid monitoring and control operations according to one embodiment of the present disclosure.
- (9) Although the following Detailed Description will proceed with reference being made to illustrative embodiments, many alternatives, modifications and variations thereof will be apparent to those skilled in the art.

DETAILED DESCRIPTION

- (10) FIG. 1 illustrates a leaching fluid control system **100** according to various embodiments of the present disclosure. The system **100** includes a flow control system **102** generally configured to monitor and control leaching fluid pressure and flow rate of leaching fluid from a leaching fluid source **103** to a controlled source manifold **105** to deliver controlled leaching fluid to a plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**. The system **100**, in some embodiments, may also include a flush control system **130** generally configured to flush the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**, via flush manifold **111**. The leaching fluid source **103** may include, for example, a pipe, tank, container, etc. that contains a leaching fluid, such as an acidic solution used to leach precious and/or industrial minerals from soil/rock. The controlled source manifold **105** and flush manifold **111** may each include a pipe, tank, container, etc. As a general matter, and depending on a particular deployment of the system **100**, the leaching fluid source **103** may be dimensioned and/or capable of delivering flow rates on the order of 100 gallons/minute. Of course, this is only an example, and the system **100** of the present disclosure may be configured to deliver any desired flow rate of leaching fluid.
- (11) The flow control system **102** includes controller circuitry **104** generally configured to control the operation of various components of the flow control system **102**, as is described in greater detail below. The flow control system **102** also includes flow rate sensor circuitry **106** in fluid communication with the leaching fluid source **103**, generally configured to sense a flow rate of

leaching fluid from the source **103** to the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**. In one example embodiment, the flow rate sensor circuitry **106** may include a flow rate sensor manufactured and distributed by Euromag (e.g., Euromag Magmeter, etc.).

(12) The flow control system **102** also includes controllable valve circuitry **108** in fluid communication with the flow rate sensor circuitry **106**. The controllable valve circuitry **108** is configured to provide a controlled flow rate of leaching fluid, to be delivered to the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**. In one example embodiment, the controllable valve circuitry **108** may include a flow control device manufactured by Euromag and distributed by Barmad Corporation. In some embodiments, the controllable valve circuitry **106** may also include pressure sensor circuitry (not shown) generally configured to determine a forward pressure from the leaching fluid source **103** and/or a back pressure from the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**. The flow rate sensor circuitry **106** is also in fluid communication with the controlled source manifold **105**, and in fluid communication with the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**.

(13) The flow control system **102** also includes power supply circuitry **110** generally configured to provide power to various components of the flow control system **102**. The power supply circuitry **110** may include one or more battery systems **112**. The battery systems **112** may include one or more rechargeable battery technologies, for example, nickle cadmium (NiCd), lithium ion (Lilon), etc. The power supply circuitry **110** may include commercially available and/or custom power supply circuit topologies, for example, a linear power supply, voltage/mode/current mode switching power supply, etc. In embodiments described herein, the power supply circuitry **110** may be coupled to a solar panel array (described below) to enable recharging of the battery system **112** using solar energy.

(14) The flow control system **102** also includes heating circuitry **114** generally configured to provide heat, directly or indirectly, to one or more components of the flow control system **102**. For example, the heating circuitry **114** may be configured to provide heat to leaching fluid flowing through the controllable valve circuitry **108** and/or the flow rate sensor circuitry **106** to prevent freezing of the leaching fluid. In some embodiments, the heating circuitry **114** may include temperature sensor circuitry **116** generally configured to sense temperature in the environment around the flow control system **102**. The controller circuitry **104** may control the operation (on, off, setting heat temperature, etc.) based on, at least in part, the temperature sensed by the temperature sensor **116**. The heating circuitry **114** may include one or more electronic heating elements (not shown) which may provide heat using power from the battery system **112**. In some embodiments, the heating circuitry **114** may also include a liquid/gas fuel supply (e.g., kerosene, propane, etc.) and one or more gas-powered heating elements to provide heat.

(15) The flow control system **102** also includes communications circuitry **118** generally configured to exchange commands and data with a remote system (described below), via network **120**. The communications circuitry **118** may communicate using a known and/or after-developed communications protocols including, for example, cellular communications protocols (e.g., LTE, 3G, 4G, 5G/6G, etc.), wireless network communications protocols (e.g., IEEE 10 BASE x, WiFi, etc.). In some embodiments, for example if the system **100** is deployed in an extremely remote location outside of cellular/wifi coverage, communications circuitry **118** may be configured to communicate using satellite communications protocols, etc. The communications circuitry **118**, in some embodiments, may also be configured exchange commands and data with the flush control system **130** (described below), via wireless communications channel **113**. The wireless communications channel **113** may include, for example, short range wireless radio frequency communications protocols (e.g., Bluetooth, etc.), infrared and/or near infrared communications protocols, wireless network communication protocols, cellular communications protocols, etc. Communications circuitry **118** may also include antennae systems (e.g., direction and/or polar antennae arrays, etc.) and/or signal boosting circuitry (not shown) to enable greater range of

communications.

(16) The controller circuitry **104** is configured to control the controllable valve circuitry **108**, based on feedback signal **109** generated by the flow rate sensor circuitry **106** and also based on one or more flow rate thresholds, e.g., **150A**, **150B**, **150C**. The one or more flow rate thresholds **150A**, **150B**, **150C** may include, for example, one or more user-definable and/or pre-programmed flow rate thresholds, which may include a nominal flow rate threshold, a maximum flow rate threshold and a minimum flow rate threshold. Initially, during operation, the controller circuitry **104** controls the controllable valve circuitry **108** to deliver a desired (e.g., nominal) flow rate of leaching fluid from the source **103** to the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**. The controller **104** may control the controllable valve circuitry **108** so that the flow rate (as measured by flow rate sensor circuitry **106**) remains within the defined minimum and maximum flow rates. Thus, the controller **104** is configured to compare the feedback signal **109** to one or more flow rate thresholds **150A**, **150B**, **150C** to determine if the flow rate is operating within a defined range. If the flow rate drops below a defined minimum flow rate, this may indicate that one or more of the leaching lines **107A**, **107B**, **107C**, . . . , **107n** may be partially or fully clogged and/or kinked at some point along the length of a leaching lines, thus restricting flow of leaching fluid. If the flow rate exceeds a defined maximum flow rate, this may indicate a leak in the systems, for example, a leak in the one or more of the leaching lines **107A**, **107B**, **107C**, . . . , **107n**. If a defined minimum and/or maximum flow rate is exceeded, the controller **104** may control the communications circuitry **118** to generate an alert message to send to the remote interface **122**, via network **120**.

(17) In some embodiments, the controller circuitry **104** may also be configured to control operations of the heating circuitry **114**, for example, based on available battery power, a desired temperature threshold, etc. Thus, the controller circuitry **104** may control the heating circuitry **114** to turn on if the temperature sensor **116** indicates the ambient temperature has dropped below a predefined and/or pre-programmed temperature threshold, and to turn off once the ambient temperature has exceeded the temperature threshold. The controller circuitry **110** may also be configured to control the operation of the power supply circuitry **110**, for example, to supply power to various components as needed. In addition, the controller circuitry **104** may also include power management circuitry (not shown) to control the charging and discharging of the battery system **112** based on, for example, power demand, available battery power, available solar power, etc. Thus, in some embodiments, the controller circuitry **104** may cause one or more components to enter into a low power (e.g., “sleep”) mode, to conserve battery power. The controller circuitry **104** may also store historical data concerning the state and status of various components (e.g., available power, temperature, flow rate alert messages, power failure alert message, heating failure alert messages, average flow rate, component status messages, time/date stamp data, etc.), which may be transmitted to the remote interface **122** on a continuous and/or periodic basis to enable remote monitoring and control of the flow control system **102**.

(18) In some deployments of the leaching fluid control system **100**, it may be advantageous to enable flushing capabilities to flush the plurality of the leaching lines **107A**, **107B**, **107C**, . . . , **107n**. Accordingly, in at least one embodiment, the leaching control system **100** may also include flush control system **130** generally configured to flush the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n**, via flush manifold **111**. The flush manifold **111** may be generally coupled to the end sections of each of the plurality of leaching lines **107A**, **107B**, **107C**, . . . , **107n** and in fluid communication with the controlled source manifold **105**. The flush control system **130** includes controllable flush valve circuitry **132** arranged in-line with respect to the flush manifold **111** and generally configured to control a flow of leaching fluid through the flush manifold **111**. The flush control system **130** may also include power supply circuitry **138** and battery circuitry **140**, similar to power supply **110**/battery **112**, described above. The flush control system **130** may also include heating circuitry **142** and temperature sensor **144**, similar to heater **114** and temperature sensor **116**, described above. To enable communication with the flow control system **102**, the flush control

system **130** may also include communications circuitry **136** to exchange commands and data with the flow control system **102**, via wireless communications channel **113**. Communications circuitry **136** may also include antennae systems (e.g., direction and/or polar antennae arrays, etc.) and/or signal boosting circuitry (not shown) to enable greater range of communications.

(19) Controller circuitry **104** may control a flushing operation by controller the flush system **130** to flush the leaching lines **107A**, **107B**, **107C**, . . . , **107n** by “opening” the leaching fluid manifold flush **111** to discharge leaching fluid. In some embodiments, flush operations may be performed at user-definable and/or programmable intervals. Thus, controller circuitry **104** may also include timer circuitry (not shown) to enable timing of flush operations. The timing circuitry may utilize a programmable timing input **152** to perform flush operations at timing intervals defined by the input **152** (for example, flush the leaching lines **107A**, **107B**, **107C**, . . . , **107n** every hour, etc.). In some embodiments, controller circuitry **104** may initiate a flush operation based on flow/pressure conditions detected in the leaching **107A**, **107B**, **107C**, . . . , **107n**. For example, a clog in one or more of the leaching lines **107A**, **107B**, **107C**, . . . , **107n** may be evident when controller circuitry **104** detects a drop in flow rate (as indicated by the flow rate sensor **106**). The controller **104** may attempt to correct the drop in pressure by opening the controllable valve **108** further to compensate for the drop in flow rate. If the flow rate remains reduced, controller **104** may initiate a flush operations by controlling the controllable flush valve circuitry **132** to flush the leaching lines **107A**, **107B**, **107C**, . . . , **107n**.

(20) After a defined time period of a flush operation (for example, 5 minutes, etc.) The system **100** may also include a remote leaching fluid monitoring/control interface **122** generally configured to exchange commands and data with the flow control system **102**, receive messages and alerts from the flow control system **102**, and to control various operational aspects of the flow control system **102**. In some embodiments, the interface **122** may be embodied as a smart phone device (e.g., iPhone, Galaxy, etc.) and/or smart tablet device (e.g., iPad, laptop computer, etc.), etc., that includes a display, communications circuitry, input circuitry (e.g., touch screen, keyboard, etc.). The interface **122** may include executable instructions and/or instruction sets, for example, in the form of an “app” or application, to perform the various task described herein. The interface **122** may include system alert(s) code **124** generally configured to trigger an alert upon receipt of an alert message from the flow control system **102**. Examples of alert messages include available power, low battery state, temperature data, flow rate alert messages, power failure alert message, heating failure alert messages, average flow rate, component status messages, time/date stamp data, etc. The alert may include, for example, flashing lights, defined sound, vibration, generation of a text and/or email message, etc., so that a user is notified of the alert message. The interface **122** may also include control actions code **126** to generate one or more control commands to control various components of the flow control system **102**. Control commands may include, for example, setting the nominal, minimum and/or maximum follow rate, initiating a flush operation, power management commands, heating commands, etc. The interface **122** may also include a monitoring database **128** to store historical data concerning the flow control system **102**.

(21) FIGS. 2A and 2B illustrate views of a housing structure **200** to contain the flow control system **102** and/or flush control system **130** according to one embodiment of the present disclosure. As illustrated in FIG. 2A, the housing **200** generally includes an outer member **202** and an inner member **204**. The inner member **204** defines a chamber to contain the flow control system **102** and/or flush control system **130**. The inner and outer members **202**, **204** may be formed of, for example, steel, plastic, etc. Insulation material **206** may be inserted between the inner member **204** and outer member **206** to reduce temperature fluctuations within the chamber. The insulation material **206** may include, for example, fiberglass insulation, filament insulation, fabric, Styrofoam, etc. A top section **208** may be separable to provide access to the chamber. As illustrated in FIG. 2B, the top section **208** may include a plurality of solar panel arrays (**210**) to provide solar electricity for the flow control system **102** and/or flush control system **130**. In some embodiments, the top

section **208** may include internal channels or conduits to enable heat to reach to the solar panel arrays (**210**) to prevent ice/snow build up on the arrays. While FIGS. **2A** and **2B** illustrate a housing structure **200** having a generally rectangular shape, it will be appreciated that the housing structure **200** may have any shape/dimensions sufficient to contain the flow control system **102** and/or flush control system **130**.

(22) FIGS. **3A** and **3B** illustrate views of a housing structure **300** to contain the flow control system **102** and/or flush control system **130** according to another embodiment of the present disclosure. As illustrated in FIG. **3A**, the housing structure **300** generally includes a container **301** defined by an outer member **302** and an inner member **304**. The inner member **304** defines a chamber to contain the flow control system **102** and/or flush control system **130**. The inner and outer members **302**, **304** may be formed of, for example, steel, plastic, etc. Insulation material **306** may be inserted between the inner member **304** and outer member **306** to reduce temperature fluctuations within the chamber. The insulation material **306** may include, for example, fiberglass insulation, filament insulation, fabric, Styrofoam, etc. A top section **308** may be separable to provide access to the chamber. While FIG. **3A** illustrate a container **301** having a generally rectangular shape, it will be appreciated that the container **301** may have any shape/dimensions sufficient to contain the flow control system **102** and/or flush control system **130**. In this embodiment, the housing structure **300** includes a boom arm **312** affixed to the container **301**. A tray member **314** is affixed to the boom arm **312** at pivot joint **316**. The tray member **314** includes a plurality of solar panel arrays **310**. The pivot joint **316** enables the tray member **314** to pivot with respect to the container **301**, as illustrated by dashed arrow **321**. The pivot joint **316** may also enable the tray member **314** to rotate with respect to the container **301**, as illustrated by dashed arrow **323**. In some embodiments, the tray member **314** may also be configured to move laterally with respect to the container, as illustrated by dashed arrow **331** (using, for example, rail sections (not shown) and/or additional boom structures, etc.). In some embodiments, the boom arm **312** may be extendable/and/or retractable to enable raising or lowering of the tray member **312** with respect to the container **301**, as illustrated by dashed arrow **333**. Enabling movement of the tray member **314** may provide for optimal positioning of the solar panels **310** to provide maximal sun exposure in a given operational environment. In some embodiments, the boom arm **312** may contain antennae structures (not shown) to enhance communication. As illustrated in FIG. **3B**, the tray member **314** may include a plurality of solar panel arrays (**310**) to provide solar electricity for the flow control system **102** and/or flush control system **130**.

(23) FIGS. **4A** and **4B** illustrate views of a leaching pad **400** according to various embodiments of the present disclosure. As is known, a leaching pad **400** is formed as earth/rock material is removed from a mine. The leaching pad **400** includes successive built up layers **402**, where the top layer **410** is generally horizontal and is leached for minerals. As layers are built-up, a slope section **414** forms having a sloped surface **408**, with an angle defined by a top point **404** and a bottom point **406**. A liner **412** is initially disposed under the leaching pad **400** to collect the slurry formed by leaching fluid percolating through the pad **410** and various layers **402**. FIG. **4B** illustrates the generally horizontal pad **410**, which may include a plurality of sections (depending on the size of the pad **410**).

(24) FIG. **5A** illustrates a deployed leaching fluid control system **500** according to one embodiment of the present disclosure. In this embodiment, the leaching lines **107A**, **107B**, **107C**, . . . , **107n** are deployed on the horizontal section of a leaching pad **410**. In this embodiment, the flow control system **102** is plumbed to minimize a distance between the leaching fluid source **103** and the controlled source manifold **105**, to maximize an area that is leached by the leaching fluid lines **107A**, **107B**, **107C**, . . . , **107n**. Each leaching line **107A**, **107B**, **107C**, . . . , **107n** includes a plurality of emitter holes/slots **109** to enable the leaching fluid to exit the leaching lines onto the soil of the leach pad **410**. The size of the holes/slots **109** and/or the number and spacing of the holes/slots **109** may be selected to deliver a desired flow rate per area onto the leach pad **410**. For

example, the size and spacing of the holes/slots **109** may be selected to deliver 0.002 gal/min/ft² of leaching fluid. Of course, this is only an example, and size and/or spacing of the holes/slots **109** may be selected to deliver any desired flow rate per area.

(25) FIG. 5B illustrates a deployed leaching fluid control system **550** according to another embodiment of the present disclosure. In this embodiment, the leaching lines **107A**, **107B**, **107C**, . . . , **107n** are deployed on the slope **408** of a leach pad **400**. Referring again to FIG. 4A, the top (rim) of the slope **408** is illustrated at **404**, and the bottom of the slope **408** is illustrated at **406**. In this embodiment, the flow control system **102** is plumbed to minimize a distance between the leaching fluid source **103** and the controlled source manifold **105**, to maximize an area that is leached by the leaching fluid lines **107A**, **107B**, **107C**, . . . , **107n**. Each leaching line **107A**, **107B**, **107C**, . . . , **107n** includes a plurality of emitter holes/slots **109** to enable the leaching fluid to exit the leaching lines onto the soil of the leach pad **410**. The size of the holes/slots **109** and/or the number and spacing of the holes/slots **109** may be selected to deliver a desired flow rate per area onto the leach pad **410**. For example, the size and spacing of the holes/slots **109** may be selected to deliver 0.002 gal/min/ft² of leaching fluid. Of course, this is only an example, and size and/or spacing of the holes/slots **109** may be selected to deliver any desired flow rate per area. As can be appreciated, placement of the leaching fluid lines **107A**, **107B**, **107C**, . . . , **107n** along the slope **408** may cause excessive pressure within the lines due to gravity. Accordingly, the leaching fluid lines **107A**, **107B**, **107C**, . . . , **107n** may each include one or more pressure regulators **115** to control pressure distribution along the length of the leaching fluid lines **107A**, **107B**, **107C**, . . . , **107n**.

(26) FIG. 6 illustrates a flowchart **600** of leaching fluid monitoring and control operations according to one embodiment of the present disclosure. Operations of this embodiment include coupling a leaching fluid control system to a leaching fluid source **602**. Operations also include coupling the leaching fluid control system to one or more leaching lines **604**. Operations also include setting a controllable valve to deliver a predetermined flow rate of leaching fluid to the leaching lines **606**. Operations also include monitoring the flow rate of the leaching fluid **608**, and determining if the flow rate is within one or more predetermined flow rate thresholds **610**. If the flow rate is within the thresholds, operations include continuing to monitor the flow rate. If the flow rate exceeds one or more flow rate thresholds, operations of this embodiment include determining if the condition is an under flow or an over flow condition **612**. Operations also include controlling the controllable valve so that the flow rate is within the flow rate thresholds **614**. Operations also include monitoring the flow rate of the leaching fluid and determining if the flow rate is within one or more predetermined flow rate thresholds **618**. If the flow rate is within the thresholds, operations include continuing to monitor the flow rate **620**. If the flow rate exceeds one or more flow rate thresholds, operations of this embodiment include controlling a flush valve to flush the leaching lines **622**. Operations also include monitoring the flow rate of the leaching fluid and determining if the flow rate is within one or more predetermined flow rate thresholds **624**. If the flow rate is within the thresholds, operations include continuing to monitor the flow rate **626**. If the flow rate exceeds one or more flow rate thresholds, operations of this embodiment include triggering a status alert message to a remote monitoring interface **628**.

(27) While FIG. 6 illustrate various operations according to one or more embodiments, it is to be understood that not all of the operations depicted in FIG. 6 are necessary for other embodiments. Indeed, it is fully contemplated herein that in other embodiments of the present disclosure, the operations depicted in FIG. 6, and/or other flowcharts and/or operations described herein, may be combined in a manner not specifically shown in any of the drawings, but still fully consistent with the present disclosure. Thus, claims directed to features and/or operations that are not exactly shown in one drawing are deemed within the scope and content of the present disclosure.

(28) As used in this application and in the claims, a list of items joined by the term “and/or” can mean any combination of the listed items. For example, the phrase “A, B and/or C” can mean A; B; C; A and B; A and C; B and C; or A, B and C. As used in this application and in the claims, a list of

items joined by the term “at least one of” can mean any combination of the listed terms. For example, the phrases “at least one of A, B or C” can mean A; B; C; A and B; A and C; B and C; or A, B and C.

(29) “Circuitry”, as used in any embodiment herein, may comprise, for example, singly or in any combination, hardwired circuitry, programmable circuitry such as processors comprising one or more individual instruction processing cores, state machine circuitry, and/or firmware that stores instructions executed by programmable circuitry. The circuitry may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), system on-chip (SoC), application-specific integrated circuit (ASIC), programmable logic devices (ILD), digital signal processors (DSP), field programmable gate array (FPGA), logic gates, registers, semiconductor device, chips, microchips, chip sets, etc. The circuitry described herein may be formed of electrical and mechanical components to operate in a manner herein described, as would be understood by one skilled in the art using conventional and/or custom components. For example, in the case of controllable valve circuitry **108** and/or **132**, fluid valve components may be formed of mechanical structures (e.g., dampers, baffles, screw valve opening/closing mechanisms, and/or other known types of mechanical valve structures, etc.) and such mechanical structures may be controlled (e.g., moved, urged, rotated, etc.) using controllable electrical components such as valve actuators and the like.

(30) Any of the operations described herein may be implemented in a system that includes one or more non-transitory storage devices having stored therein, individually or in combination, instructions that when executed by circuitry perform the operations. The storage device includes any type of tangible medium, for example, any type of disk including hard disks, floppy disks, optical disks, compact disk read-only memories (CD-ROMs), compact disk rewritables (CD-RWs), and magneto-optical disks, semiconductor devices such as read-only memories (ROMs), random access memories (RAMs) such as dynamic and static RAMs, erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), flash memories, Solid State Disks (SSDs), embedded multimedia cards (eMMCs), secure digital input/output (SDIO) cards, magnetic or optical cards, or any type of media suitable for storing electronic instructions. Other embodiments may be implemented as software executed by a programmable control device. Also, it is intended that operations described herein may be distributed across a plurality of physical devices, such as processing structures at more than one different physical location.

(31) The terms and expressions which have been employed herein are used as terms of description and not of limitation, and there is no intention, in the use of such terms and expressions, of excluding any equivalents of the features shown and described (or portions thereof), and it is recognized that various modifications are possible within the scope of the claims. Accordingly, the claims are intended to cover all such equivalents. Various features, aspects, and embodiments have been described herein. The features, aspects, and embodiments are susceptible to combination with one another as well as to variation and modification, as will be understood by those having skill in the art. The present disclosure should, therefore, be considered to encompass such combinations, variations, and modifications.

(32) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

Claims

1. A leaching fluid control system for mining operations, comprising: controllable valve circuitry in fluid communication with a leaching fluid source, the controllable valve circuitry to provide a controlled flow rate of the leaching fluid; flow rate sensor circuitry in fluid communication with the controllable valve circuitry and with one or more leaching fluid lines to generate a flow rate signal indicative of the leaching fluid flow rate through the controllable valve circuitry; controller circuitry to control the controllable valve circuitry to deliver a flow rate of leaching fluid based on the flow rate signal and one or more flow rate thresholds; the controller circuitry also to determine, based on the flow rate signal, if the flow rate drops below a minimum flow rate threshold indicative of a clogged and/or kinked condition of the one or more leaching fluid lines; and to determine, based on the flow rate signal, if the fluid flow rate exceeds a maximum flow rate threshold indicative of a leak associated with one or more of the leaching fluid lines; wherein the leaching fluid lines are disposed vertically on a slope section of a leach pad; and a flush manifold coupled to the leaching fluid lines and a controllable flush valve to flush the leaching lines; wherein the controller circuitry is configured to control the controllable flush valve to begin a flush operation if the flow rate drops below the minimum flow rate threshold; and wherein the controller circuitry further to trigger a status alert message to a remote monitoring interface if the fluid flow rate exceeds the maximum flow rate threshold.
2. The system of claim 1, further comprising power supply circuitry and a rechargeable battery system to provide power to at least the controller circuitry and the controllable valve circuitry.
3. The system of claim 2, further comprising a solar panel array to provide solar electricity to the power supply circuitry to recharge the rechargeable battery system.
4. The system of claim 1, further comprising heating circuitry to provide a heating source to at least the controllable valve circuitry.
5. The system of claim 1, further comprising communications circuitry to exchange commands and data with a remote interface.
6. A leaching fluid control system for mining operations, comprising: controllable valve circuitry in fluid communication with a leaching fluid source, the controllable valve circuitry to provide a controlled flow rate of the leaching fluid; flow rate sensor circuitry in fluid communication with the controllable valve circuitry and with one or more leaching fluid lines to generate a flow rate signal indicative of the leaching fluid flow rate through the controllable valve circuitry; controller circuitry to control the controllable valve circuitry to deliver a flow rate of leaching fluid based on the flow rate signal and one or more flow rate thresholds; the controller circuitry also to determine, based on the flow rate signal, if the flow rate drops below a minimum flow rate threshold indicative of a clogged and/or kinked condition of the one or more leaching fluid lines; and to determine, based on the flow rate signal, if the fluid flow rate exceeds a maximum flow rate threshold indicative of a leak associated with one or more of the leaching fluid lines; and a flush manifold coupled to the leaching fluid lines and a controllable flush valve to flush the leaching lines; wherein the controller circuitry is configured to control the controllable flush valve to begin a flush operation if the flow rate drops below the minimum flow rate threshold; and wherein the controller circuitry further to trigger a status alert message to a remote monitoring interface if the fluid flow rate exceeds the maximum flow rate threshold.
7. The system of claim 6, further comprising power supply circuitry and a rechargeable battery system to provide power to at least the controller circuitry and the controllable valve circuitry.
8. The system of claim 7, further comprising a solar panel array to provide solar electricity to the power supply circuitry to recharge the rechargeable battery system.
9. The system of claim 6, further comprising heating circuitry to provide a heating source to at least the controllable valve circuitry.
10. The system of claim 6, further comprising communications circuitry to exchange commands and data with a remote interface.

11. The system of claim 6, wherein the leaching fluid lines are disposed vertically on a slope section of a leach pad.

12. The system of claim 6, wherein the leaching fluid lines are disposed horizontally on a horizontal section of a leach pad.
